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Multidrug-resistant *Klebsiella pneumoniae* clinical isolates producing NDM- and OXA-type carbapenemase in NepalSatomi Takei^a, Yoko Tabe^a, Takashi Miida^a, Tomomi Hishinuma^b, Abdullah Khasawneh^a, Teruo Kirikae^c, Jeevan B. Sherchand^d, Tatsuya Tada^{b,*}^a Department of Clinical Laboratory Medicine, Juntendo University Graduate School of Medicine, Tokyo, Japan^b Department of Microbiology, Juntendo University Graduate School of Medicine, Tokyo, Japan^c Department of Microbiome Research, Juntendo University Graduate School of Medicine, Tokyo, Japan^d Department of Medical Microbiology, Tribhuvan University, Maharajgunj, Kathmandu, Nepal

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ABSTRACT

Objectives: The emergence of multidrug-resistant *Klebsiella pneumoniae* has become a serious problem in medical settings worldwide.**Methods:** A total of 46 isolates of multidrug-resistant *K. pneumoniae* were obtained from 2 hospitals in Nepal from October 2018 to April 2019.**Results:** Most of these isolates were highly resistant to carbapenems, aminoglycosides, and fluoroquinolones with the minimum inhibitory concentrations (MICs) of more than 64 μ g/mL. These isolates harboured carbapenemase-encoding genes, including *bla*_{NDM-1}, *bla*_{NDM-5}, *bla*_{OXA-181} and *bla*_{OXA-232}, and 16S rRNA methyltransferase-encoding genes, including *armA*, *rmtB*, *rmtC*, and *rmtF*. Multilocus sequence typing revealed that 44 of 46 isolates were high-risk clones such as ST11 (2%), ST14 (4%), ST15 (11%), ST37 (2%), ST101 (2%), ST147 (28%), ST231 (13%), ST340 (4%), and ST395 (28%). In particular, ST395 isolates, which spread across medical settings in Nepal, co-harboured *bla*_{NDM-5} and *rmtB* on IncFII plasmids and co-harboured *bla*_{OXA-181/232} and *rmtF* on ColKP3 plasmids. Several isolates harboured *bla*_{OXA-181} or *bla*_{NDM-5} on their chromosomes and multi-copies of *bla*_{NDM-1} or genes encoding 16S rRNA methyltransferases on their plasmids.**Conclusions:** The presented study demonstrates that the high-risk clones of multidrug-resistant *K. pneumoniae* spread in a clonal manner across hospitals in Nepal.

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1. Introduction

The emergence and spread of multidrug-resistant (MDR) *Enterobacteriales* has become global concern [1]. Most of these isolates produce carbapenemases, including imipenemase (IMP), *Klebsiella pneumoniae* carbapenemase (KPC), New Delhi metallo- β -lactamase (NDM), and oxacillinase (OXA)-48-likes. IMP, KPC, NDM, and OXA-48-likes were first detected from a patient infected with *Serratia marcescens* in Japan in 1991, *K. pneumoniae* in the United States in 1996, *Escherichia coli* and *K. pneumoniae* in Sweden in 2008, and *K. pneumoniae* in Turkey in 2001, respectively [2,3]. To date, 96 vari-

ants of IMPs, 184 variants of KPCs, 59 variants of NDMs and 27 variants of OXA-48-likes were detected from clinical and environmental isolates (https://www.ncbi.nlm.nih.gov/pathogens/refgene/#gene_family). Carbapenemase-encoding genes in *Enterobacteriales* are frequently located on plasmids belonging to various incompatibility (Inc) types.

Most MDR *K. pneumoniae* isolates resistant to aminoglycosides produce aminoglycoside *N*-acetyltransferases and aminoglycoside *O*-nucleotidyltransferases [4]. In addition, the increasing numbers of MDR *K. pneumoniae* isolates have been reported to produce 16S rRNA methyltransferases, which are extremely high resistance to most aminoglycosides [5]. To date, 11 classes of 16S rRNA methyltransferases, including ArmA, RmtA to H, NpmA and B, have been detected from clinical isolates worldwide [6,7].

Fluoroquinolone resistance has been associated with chromosomal-mediated mutations in the quinolone resistance-

* Corresponding author: Tatsuya Tada, Department of Microbiology, Juntendo University Graduate School of Medicine, 2-1-1 Hongo, Bunkyo-ku, Tokyo 113-8421, Japan. Tel.: (81) 3 3813 3111; fax: (81) 3 5684 7830.

E-mail address: t-tada@juntendo.ac.jp (T. Tada).

determining region, which includes the *gyrA* and *parC* genes that encode DNA gyrase and topoisomerase IV, respectively [8]. The plasmid-mediated genes including *qnrB1* or *qnrS1* were firstly detected in quinolone-resistant *Shigella flexneri* reported in Japan in 2003 and *K. pneumoniae* in India in 2002, respectively [9,10]. These plasmid-mediated quinolone resistant factors could contribute to the development of higher-level fluoroquinolone resistance and rapid spread of resistance among *Enterobacteriales*.

The current situation of drug-resistant *Enterobacteriales* in medical settings in Nepal is unclear. In this study, we describe the molecular epidemiology of the multidrug resistant isolates of *K. pneumoniae* obtained from patients in Nepal from October 2018 to April 2019.

2. Materials and methods

2.1. Bacterial strains

In 92 isolates screened by EUCAST disk diffusion test method (https://www.eucast.org/ast_of_bacteria/disk_diffusion_methodology; last accessed 22 February 2024), 46 clinical isolates of multidrug-resistant *K. pneumoniae* complex, defined as strains showing resistance to imipenem or meropenem (MIC, ≥ 4 $\mu\text{g/mL}$), amikacin (MIC, ≥ 64 $\mu\text{g/mL}$), and ciprofloxacin (MIC, ≥ 1 $\mu\text{g/mL}$), were obtained between October 2018 and April 2019 from patients treated at 2 hospitals in Nepal. Of these 46 isolates, 10 were from hospitals A and 36 were from hospital B. Of the 46 isolates, 28 isolates were obtained from sputum, 11 from urine, 6 from pus and wounds, and 1 from blood.

2.2. Antimicrobial susceptibility testing

Antimicrobial susceptibilities were tested as described by the guidelines of the Clinical and Laboratory Standard Institute (CLSI) [11]. The ranges of antibiotic concentrations tested were amikacin (AMK) 0.5 to 512 $\mu\text{g/mL}$, aztreonam (AZT) 0.5 to 512 $\mu\text{g/mL}$, ceftazidime (CAZ) 0.5 to 512 $\mu\text{g/mL}$, ciprofloxacin (CIP) 0.25 to 512 $\mu\text{g/mL}$, colistin (CST) 0.0625 to 128 $\mu\text{g/mL}$, imipenem (IPM) 0.5 to 512 $\mu\text{g/mL}$, meropenem (MEM) 0.5 to 512 $\mu\text{g/mL}$, and tigecycline (TGC) 0.5 to 512 $\mu\text{g/mL}$. MICs of each antimicrobial agent were determined by broth microdilution methods using Muller Hinton broth and 96-well microtiter plates (Kohjin Bio, Co., Ltd. Saitama, Japan).

2.3. Whole genome sequencing and genomic analysis

Genomic DNAs of the 46 isolates were extracted using DNeasy blood and tissue kits (Qiagen, Tokyo, Japan) or G1AGEN genomic tip 20/G (Qiagen), and their complete genomes were sequenced using Miseq platform (Illumina, San Diego, CA) and MinION (Oxford Nanopore Technologies, Oxford, United Kingdom). Raw reads of each isolate were assembled using CLC Genomic Workbench version 10.0.1 (CLC bio, Aarhus, Denmark). Species identities of these isolates were determined using an ANI calculator [12] or a Type (Strain) Genome Server (TYGS) (<https://tygs.dsmz.de>). The sequences of antimicrobial resistance genes were determined using ResFinder 4.1 and plasmids were typed using Plasmid finder 2.1, both from the Centre for Genomic Epidemiology (CGE) (<https://www.genomicepidemiology.org/>). The sequences of plasmids were annotated using DDBJ Fast Annotation and Submission Tool (<https://dfast.ddbj.nig.ac.jp>). The mutations of quinolone resistance-determining regions, which includes the *gyrA* and *parC* genes [13], were detected in silico using CLC Genomics Workbench v.11.0.1 (CLC Bio, Denmark). Comparative analysis of sequences surrounding carbapenemase- and 16S rRNA methyltransferase-encoding genes was performed using BLAST and

visualized using In Silico Molecular Cloning (In Silico Biology Inc., Kanagawa, Japan). The images of plasmids or chromosomes were performed using proksee (<https://proksee.ca/>).

2.4. MLST, serotypes, and phylogenetic analysis

Multilocus sequence typing (MLST) was determined as described by the protocols in the MLST databases (<https://bigsd.b.pasteur.fr/>). Capsular polysaccharide (K) serotypes were determined using Kaptive ver.2.0.6 (<https://github.com/katholt/kaptive>). Phylogenetic trees were constructed using kSNP4 software based on the pan-genome SNPs (<https://sourceforge.net/projects/ksnp/>) [14] and visualized using FigTree ver.1.4.4 (<http://tree.bio.ed.ac.uk/software/figtree/>). The type strain of *K. pneumoniae* subsp. *pneumoniae* NCTC 9633 was used as the reference strain. The single nucleotide polymorphism (SNP) matrix was calculated using snp-dists ver.0.8.2 (<https://github.com/tseemann/snp-dists>).

2.5. Accession number(s)

The whole-genome sequences of all 46 isolates have been deposited in GenBank as accession no. [DRA017548](#), [LC506716](#), [LC506717](#), [LC506718](#), [LC506720](#), [LC506719](#), and [LC507653](#).

3. Results

3.1. Antimicrobial susceptibility profiles of multidrug-resistant *K. pneumoniae*

The whole-genome sequences revealed that all 46 multidrug-resistant *K. pneumoniae* complex isolates were *K. pneumoniae* subsp. *pneumoniae*. The 46 isolates were resistant to amikacin, aztreonam, ceftazidime, ciprofloxacin, imipenem, meropenem, and tigecycline, whereas 44 of 46 isolates were susceptible to colistin (Table 1). Furthermore, the isolates were extremely high resistant to amikacin, with MICs >512 $\mu\text{g/mL}$.

3.2. Antimicrobial resistance genes of multidrug-resistant *K. pneumoniae* complex

All 46 isolates of *K. pneumoniae* subsp. *pneumoniae* harboured carbapenemase encoding genes, including *bla_{NDM}* and/or *bla_{OXA-48-like}*. Of them, 14 harboured both *bla_{NDM-5}* and *bla_{OXA-232}*, 7 had both *bla_{NDM-5}* and *bla_{OXA-181}*, 7 had *bla_{NDM-1}*, 6 had *bla_{NDM-5}*, 5 had both *bla_{NDM-1}* and *bla_{OXA-232}*, 5 had *bla_{OXA-232}*, and 2 had *bla_{OXA-181}*. All isolates harboured genes encoding 16S rRNA methyltransferases, including *armA*, *rmtB*, *rmtC*, and/or *rmtF*, which contribute to high aminoglycoside resistance. Forty-four isolates (96%) harboured *aac(6')-Ib-cr*, which is the most common plasmid-mediated aminoglycoside/quinolone resistance gene (6). The 45 isolates (98%) had point mutations at quinolone resistance determining regions, including *GyrA* and *ParC* and 41 isolates (89%) had *qnrB1* and/or *qnrS1* gene(s) (Table 2).

3.3. MLST, serotypes, and phylogenetic analyses of multidrug-resistant *K. pneumoniae* isolates

Multilocus sequence typing (MLST) analysis revealed that 13 isolates (28%) belonged to sequence type 147 (ST147), 13 isolates (28%) to ST395, 6 (13%) to ST231, 5 (11%) to ST15, 2 (4%) to ST14, 2 (4%) to ST340, and each of one (2%) to ST11, ST37, ST101, ST3392, and ST4843. A phylogenetic tree of the whole-genome revealed that 45 of 46 isolates belonged to three clades, A, B, and C (Fig. 1). Clade A consisted of ST340 isolates, clade B consisted of ST395, and clade C consisted of ST3 (reference strain), ST14, ST15, ST37, ST101, ST147, ST231, ST3392, and ST4843 (Fig. 1). Of the all 46 isolates

Table 1
Antimicrobial susceptibility profiles of multidrug-resistant *K. pneumoniae* isolates in Nepal (N = 46).

Antibiotics reagent	Breakpoint for resistance ^a (μg/mL)	No. of resistant isolates (%)	MIC (μg/mL)		
			Range	MIC ₅₀	MIC ₉₀
Amikacin	≥64	100	>512	>512	>512
Aztreonam	≥16	100	32→512	512	>512
Ceftazidime	≥16	100	32→512	>512	>512
Ciprofloxacin	≥1	100	8→512	128	>512
Colistin	≥4	4	0.0625→128	0.0625	2
Imipenem	≥4	100	4–256	64	128
Meropenem	≥4	100	16–512	256	256
Tigecycline ^b	≥0.5	100	0.5–4	1	2

^a Breakpoints for antimicrobial resistance were determined according to CLSI guidelines.
^b Breakpoint for tigecycline was determined according to EUCAST guidelines.

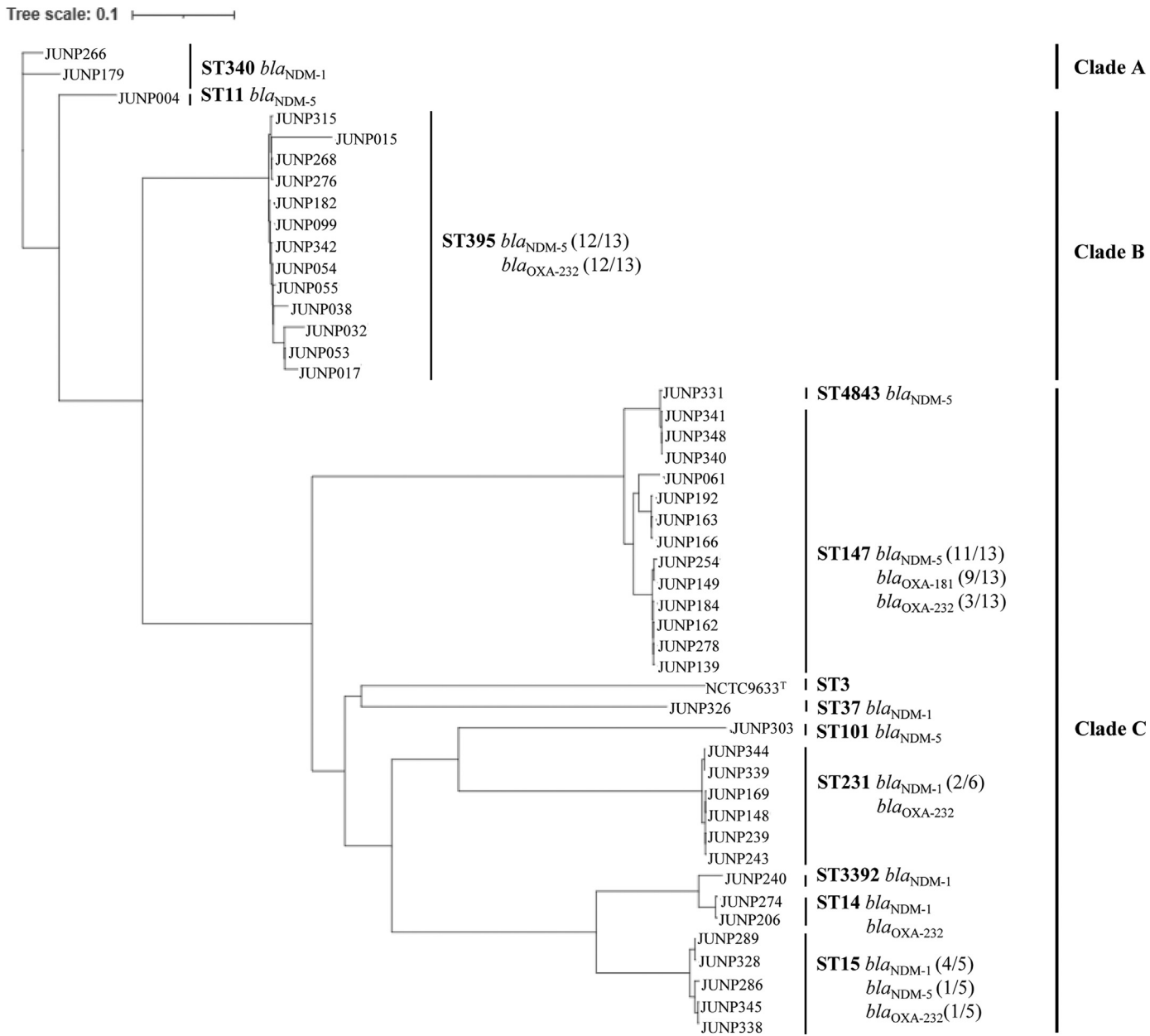


Fig. 1. Phylogenetic tree of 46 isolates of multidrug-resistant *K. pneumoniae* obtained from clinical samples in Nepal. The tree revealed three clades. Clade A included ST340, clade B included ST395, and clade C included ST3, ST14, ST15, ST37, ST101, ST147, ST231, ST3392, and ST4843. The tree was constructed by the maximum-likelihood method based on the pan-genome single nucleotide polymorphisms (SNPs).

Table 2Summary of the characteristics of the 46 multidrug-resistant *K. pneumoniae* strains, including MLSTs and drug resistance genes.

MLST	Number of isolates	Hospitals	Carbapenemase genes(s)	Extended spectrum β -lactamase encoding gene(s)	16S rRNA methyltransferase gene(s)	Aminoglycoside resistance gene(s)	Mutation(s) in DNA gyrase		Plasmid-mediated quinolone resistance protein genes	Serotype (s)
							GyrA	ParC		
ST11	1	A	<i>bla</i> _{NDM-5}	<i>bla</i> _{CTX-M-15}	<i>rmtB</i>	<i>aac</i> (3)- <i>Ila</i> , <i>aac</i> (6')- <i>Ib-cr</i> , <i>aadA2</i>	S83F, D87A	S80I	–	K24
ST14	2	B	<i>bla</i> _{NDM-1} , <i>bla</i> _{OXA-232}	<i>bla</i> _{CTX-M-15}	<i>armA</i>	<i>aac</i> (3)- <i>lid</i> (1/2), <i>aac</i> (6')- <i>Ib-cr</i> , <i>aadA1</i> (1/2), <i>aadA2</i>	S83Y, D87G	S80I	<i>qnrB1</i>	K2
ST15	5	B	<i>bla</i> _{NDM-1} (4/5), <i>bla</i> _{NDM-5} (1/5), <i>bla</i> _{OXA-232} (1/5)	<i>bla</i> _{CTX-M-15}	<i>rmtB</i> (1/5), <i>rmtC</i> (4/5), <i>rmtF</i> (3/5)	<i>aac</i> (3)- <i>lia</i> (2/5), <i>aac</i> (3)- <i>lid</i> (2/5), <i>aac</i> (6')- <i>Ib-cr</i> , <i>aadA1</i> (2/5), <i>aadA2</i> (2/5)	S83F, D87A	S80I	<i>qnrB1</i> (3/5)	K2 (2/5), K60 (3/5)
ST37	1	B	<i>bla</i> _{NDM-1}	<i>bla</i> _{CTX-M-15} , <i>bla</i> _{SHV-40} , <i>bla</i> _{SHV-56} , <i>bla</i> _{SHV-79} , <i>bla</i> _{SHV-85} , <i>bla</i> _{SHV-89}	<i>armA</i>	<i>aac</i> (3)- <i>lia</i> , <i>aac</i> (3)- <i>lid</i> , <i>aac</i> (6')- <i>Ib-cr</i> , <i>aadA1</i>	–	–	<i>qnrB1</i>	K15
ST101	1	B	<i>bla</i> _{NDM-5}	<i>bla</i> _{CTX-M-15}	<i>rmtB</i>	<i>aac</i> (6')- <i>Ib-cr</i> , <i>aadA2</i>	S83Y, D87G	S80I	–	K17
ST147	13	A (1/13), B (12/13)	<i>bla</i> _{NDM-5} (11/13), <i>bla</i> _{OXA-181} (9/13), <i>bla</i> _{OXA-232} (3/13)	<i>bla</i> _{CTX-M-15} (12/13), <i>bla</i> _{SHV-11} , <i>bla</i> _{TEM-33} (2/13), <i>bla</i> _{TEM-34} (2/13), <i>bla</i> _{TEM-141} (2/13)	<i>armA</i> (4/13), <i>rmtB</i> (10/13), <i>rmtF</i> (9/13)	<i>aac</i> (6')- <i>Ib-cr</i> (11/13), <i>aadA1</i> (4/13), <i>aadA2</i> (11/13), <i>aadA24</i> (1/13)	S83I (9/13), S83Y (4/13), D87A (4/13)	S80I	<i>qnrB1</i>	K10 (4/13), K51 (6/13), K64 (3/13)
ST231	6	B	<i>bla</i> _{NDM-1} (2/6), <i>bla</i> _{OXA-232}	<i>bla</i> _{CTX-M-15} , <i>bla</i> _{SHV-28} (1/6), <i>bla</i> _{TEM-213} (1/6)	<i>rmtF</i>	<i>aac</i> (6')- <i>Ib-cr</i> , <i>aadA1</i> (3/6), <i>aadA2</i>	S83I	S80I	<i>qnrS1</i>	K51
ST340	2	B	<i>bla</i> _{NDM-1}	<i>bla</i> _{CTX-M-15}	<i>rmtC</i> (1/2), <i>rmtF</i>	<i>aac</i> (3)- <i>lia</i> (1/2), <i>aac</i> (6')- <i>Ib-cr</i>	S83I	S80I	<i>qnrB1</i> (1/2), <i>qnrS1</i> (1/2)	K64 (1/2), unknown (1/2)
ST395	13	A (8/13), B (5/13)	<i>bla</i> _{NDM-5} (12/13), <i>bla</i> _{OXA-232} (12/13)	<i>bla</i> _{CTX-M-15} , <i>bla</i> _{CTX-M-33} (1/13), <i>bla</i> _{SHV-11} (4/13), <i>bla</i> _{SHV-12} (2/13), <i>bla</i> _{SHV-107} (1/13), <i>bla</i> _{SHV-182} (1/13)	<i>rmtB</i>	<i>aac</i> (3)- <i>lid</i> (12/13), <i>aac</i> (6')- <i>Ib-cr</i> , <i>aadA2</i> (12/13)	S83I (11/13), S83F (2/13), D87A (2/13)	S80I	<i>qnrB1</i> (12/13), <i>qnrS1</i>	K64
ST3392	1	B	<i>bla</i> _{NDM-1}	<i>bla</i> _{CTX-M-15}	<i>armA</i> , <i>rmtF</i>	<i>aac</i> (6')- <i>Ib-cr</i> , <i>aadA1</i>	S83Y, D87G	S80I	<i>qnrS1</i>	K64
ST4843	1	B	<i>bla</i> _{NDM-5}	<i>bla</i> _{CTX-M-15} , <i>bla</i> _{SHV-11}	<i>rmtB</i>	<i>aac</i> (6')- <i>Ib-cr</i> , <i>aadA2</i>	S83I	S80I	–	K64

Table 3
Genetic characterization of plasmids in multidrug-resistant *K. pneumoniae* isolates.

Number of isolates	Hospitals	MLST	Plasmid type	Size (bp)	Carbapenemase- and extended spectrum β -lactamase encoding gene(s)	Aminoglycoside resistance genes
25	A (9/25), B (16/25)	ST11(1/25), ST15 (1/25), ST147 (11/25), ST395 (11/25), ST4843 (1/25)	IncFII (16/25) ColKIP3 (1/25)	82 799–113 392 6141–12 195	<i>bla</i> _{NDM-5} , <i>bla</i> _{TEM-33} (1/25), <i>bla</i> _{TEM-34} (1/25), <i>bla</i> _{TEM-141} (1/25) <i>bla</i> _{OXA-181} (2/25) <i>bla</i> _{OXA-232} (14/25)	<i>rmtB</i> (24/25), <i>rmtF</i> (1/25), <i>aadA2</i> (22/25), <i>aac</i> (6')-Ib-cr (1/25) –
6	A (1/6), B (5/6)	ST147 (1/6), ST231(4/6), ST395 (1/6)	ColKIP3	6141–6812	<i>bla</i> _{OXA-181} (1/6) <i>bla</i> _{OXA-232} (5/6)	–
3	B	ST231(2/3), ST3392 (1/3)	IncFIB(pQil)/IncFII(K)	136 199	<i>bla</i> _{NDM-1} , <i>bla</i> _{CTX-M-15}	<i>rmtF</i> , <i>aac</i> (6')-Ib-cr, <i>aadA1</i> –
2	B	ST14	ColKIP3 (2/3) IncFIB(pNDM-Mar)/IncHI1B(pNDM-MAR)	6141 305 049– 306 968	<i>bla</i> _{OXA-232} <i>bla</i> _{NDM-1} , <i>bla</i> _{CTX-M-15}	<i>armA</i> , <i>aac</i> (6')-Ib-cr, <i>aadA2</i> –
2	B	ST15	ColKIP3 IncC	6141 156 977	<i>bla</i> _{OXA-232} <i>bla</i> _{NDM-1}	<i>rmtC</i> , <i>rmtF</i> , <i>aac</i> (6')-Ib-cr –
1	B	ST15	ColKIP3 (1/2) IncC/IncFIB(pNDM-Mar)/IncHI1B(pNDM-MAR)	6141 399 616	<i>bla</i> _{OXA-232} <i>bla</i> _{NDM-1} (2 copies), <i>bla</i> _{CTX-M-15}	<i>rmtC</i> , <i>aac</i> (3)-Iia, <i>aac</i> (3)-Iid, <i>aac</i> (6')-Ib-cr, <i>aadA2</i> <i>aadA2</i> <i>rmtC</i> , <i>aac</i> (6')-Ib-cr <i>aadA2</i>
1	B	ST15	IncFII IncC	82 833 137 575	<i>bla</i> _{NDM-1} <i>bla</i> _{NDM-1}	<i>rmtC</i> , <i>aac</i> (6')-Ib-cr <i>aadA2</i>
1	B	ST37	IncFII IncM2	82 869 88 278	<i>bla</i> _{NDM-1} <i>bla</i> _{NDM-1}	<i>armA</i> , <i>aac</i> (3)-Iid <i>rmtF</i> , <i>aac</i> (6')-Ib-cr
1	B	ST340	IncFII(K)	126 649	<i>bla</i> _{NDM-1}	<i>rmtC</i>
1	B	ST340	IncFII(Yp)	112 137	<i>bla</i> _{NDM-1}	<i>rmtB</i> , <i>aac</i> (3)-Iid, <i>aac</i> (6')-Ib-cr, <i>aadA2</i>
1	B	ST395	IncFIB(pQil)	97 394	<i>bla</i> _{NDM-5} , <i>bla</i> _{CTX-M-15}	

tested, 44 isolates were high-risk clonal lineages including ST11, ST14, ST15, ST37, ST101, ST147, ST231, ST340, and ST395.

Of 13 ST147 isolates, 4 isolates belonged to capsular polysaccharide K10, 6 to K51 and 4 to K64. All of 13 ST395 isolates belonged to K64 (Table 2).

The SNPs range of the 13 ST147 isolates from hospital A and B were from 47 to 2577 (Supplemental Table 1) and the 13 ST395 isolates from hospital A and B were from 35 to 3108 (Supplemental Table 2), revealing that several isolates belonging to ST147 and ST395 were clonally disseminating in hospital B.

3.4. Plasmids carrying *bla*_{NDM-1/–5} or *bla*_{OXA-181/–232}

Whole genome sequencing showed that 38 isolates contained plasmids carrying *bla*_{NDM-1} or *bla*_{NDM-5} ranging in size from 82,799 bp to 399,619 bp. These plasmids belonged to nine types of plasmid incompatibility complexes, including IncFII (27 isolates), IncC (3 isolates), IncFIB(pQil)/IncFII(K) (3 isolates), IncFIB(pNDM-Mar)/IncHI1B(pNDM-MAR) (2 isolates), IncC/IncFIB(pNDM-Mar)/IncHI1B(pNDM-MAR) (1 isolate), IncFII (K) (1 isolate), IncFII (Yp) (1 isolate), IncFIB (pQil) (1 isolate) and IncM2 (1 isolate) (Table 3). The *bla*_{NDM-1} genes were located on IncC, IncC/IncFIB(pNDM-Mar)/IncHI1B(pNDM-MAR), IncFIB(pNDM-Mar)/IncHI1B(pNDM-MAR), IncFIB(pQil)/IncFII(K), IncFII, IncFII(K), IncFII(Yp), and IncM2-type plasmids. The *bla*_{NDM-5} genes were located on IncFIB(pQil) and IncFII plasmids (Table 3). All the 38 isolates harbouring *bla*_{NDM-5} also harboured 16S rRNA methyltransferase-encoding genes, including *armA*, *rmtB*, *rmtC* and *rmtF*. The isolate JUNP289 harboured 3 copies of *bla*_{NDM-1}. A copy of *bla*_{NDM-1} was located on IncFII and the remaining two copies were located on IncC/IncFIB(pNDM-Mar)/IncHI1B(pNDM-MAR). The isolate JUNP328 harboured 2 copies of *bla*_{NDM-1} located on IncC and IncFII, respectively (Fig. 2).

Whole genome sequencing showed that 27 isolates contained plasmids carrying *bla*_{OXA-181} or *bla*_{OXA-232} ranging in size from 6141 bp to 12 195 bp. All plasmids belonged to ColKIP3. Of the 27

plasmids carrying *bla*_{OXA-181}, 21 (78%) harboured genes encoding both NDMs and OXAs.

A phylogenetic tree of plasmids carrying *bla*_{NDM-5} revealed that the majority of the plasmids belonged to IncFII (Fig. S1). Nine of 27 plasmids belonging to IncFII had identical structures. All nine plasmids were detected in ST395 *K. pneumoniae* isolated in hospitals A and B.

3.5. Genetic environments surrounding *bla*_{NDM-1/–5} or *bla*_{OXA-181/–232} in plasmids

The structures of the genomic environments surrounding *bla*_{NDM-1}, *bla*_{NDM-5}, *bla*_{OXA-181}, and *bla*_{OXA-232} are shown in Fig. 3. The genetic structure surrounding *bla*_{NDM-1} had 9 types (Fig. 3A–I), *bla*_{NDM-5} had 4 types (Fig. 3J–M), *bla*_{OXA-181} had 1 type (Fig. 3N) and *bla*_{OXA-232} had 2 types (Fig. 3O and P).

The structure of type A was *orf-pemI-pemK-IS1A-Tn3-IS26-IS30-bla*_{NDM-1}-*ble*_{MBL}-*trpF-tat-ISCRI-sul1-qacEΔ*, which was identical to the structures of plasmids of *K. pneumoniae* p_b78b_NDM5 in Bangladesh in 2017 (GenBank accession no. CP095588). The structure of type B was *copG-dnmt1-nth-rmtC-IS1A-bla*_{NDM-1}-*ble*_{MBL}-*trpF-dsbD-cutA-groES-groL-orf*, which was identical to the structures of plasmids of *K. pneumoniae* Carbapenemase (NDM-1)_IncA/C2 in Hong Kong in 2017 (accession no. CP050164). The structure of type C was *Tn3-orf-IS26-orf-bla*_{NDM-1}-*ble*_{MBL}-*trpF-tat-cutA-IS26-orf*, which was unique. The structure of type D was *IS15DI-IS3-aph(3')-VI-tnp1062-ISEc33-bla*_{NDM-1}-*ble*_{MBL}-*trpF-tat-cutA-groES-groL-ISCRI-2-orf-ISAbi125*, which was identical to the structures of plasmids of *K. pneumoniae* pKPN528-1 in the United States in 2012 (accession no. CP020854). The structure of type E was *orf-copG-dnmt1-nth-rmtC-ISA1-bla*_{NDM-1}-*ble*_{MBL}-*trpF-dsbD-ISCRI-sul1-qacEΔ-aadA2-orf*, which was unique. The structure of type F was *IS26-catB-bla*_{OXA-1}-*aac*(6')-Ib-cr-IS26-IS30-*bla*_{NDM-1}-*ble*_{MBL}-*trpF-tat-cutA-groES-groL-ISCRI-rhsI*, which was unique. The structure of type G was *orf-IS5057-tnpA-IS30-bla*_{NDM-1}-*ble*_{MBL}-*trpF-tat-cutA-groES-groL-tnpA-orf*, which was identical to the

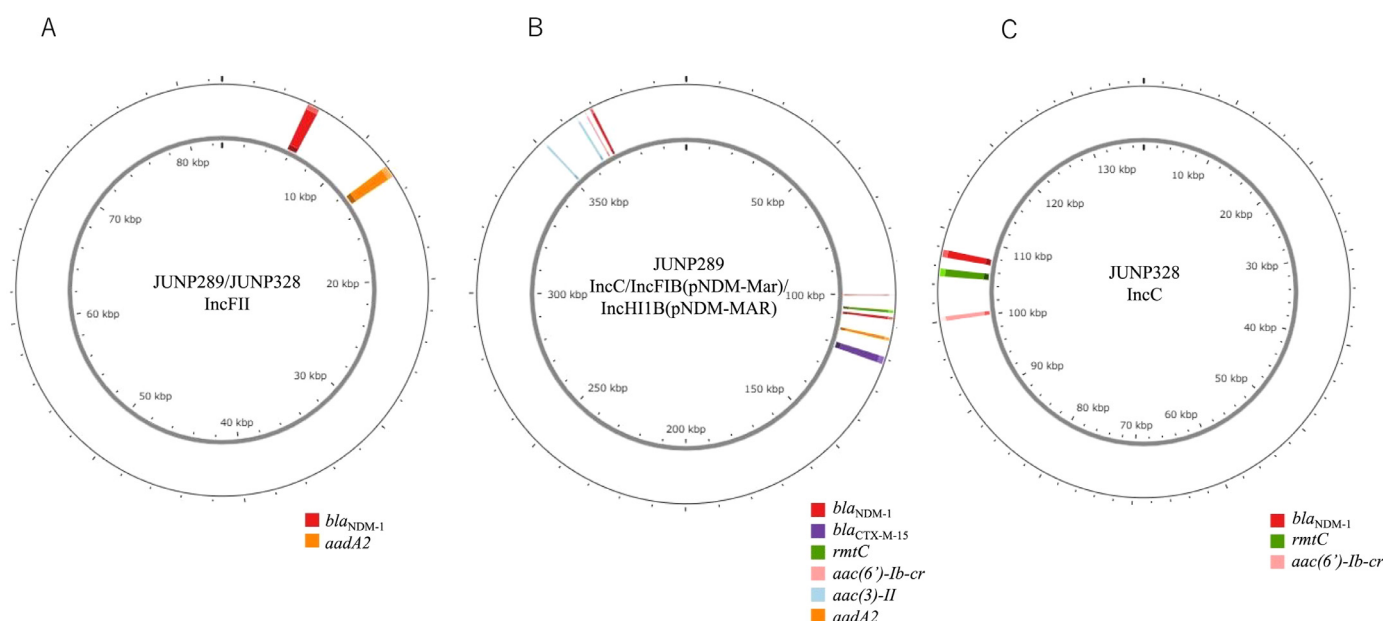


Fig. 2. The plasmid structures of JUNP289 and JUNP328 harbouring multi-copies of *bla*_{NDM-1}. (A) IncFII plasmids in JUNP289 and JUNP328 carried a copy of *bla*_{NDM-1}. (B) IncC/IncFIB(pNDM-Mar)/IncHI1B(pNDM-MAR) plasmid in JUNP289 carried two copies of *bla*_{NDM-1}. (C) IncC plasmid in JUNP328 carried a copy of *bla*_{NDM-1}. The images were generated using proksee (<https://proksee.ca/>).

structures of plasmids of *K. pneumoniae* p_dm883b_NDM1 in Bangladesh in 2017 (accession no. [CP095661](#)). The structure of type H was *tniQ-tniB-tnsB-ISCR2-rmtC-bla*_{NDM-1}-*ble*_{MBL}-*trpF-dsbD-cutA-groES-groL-ISCR2-IS30-IS3-IS5*, which was identical to the structure of plasmids of *K. pneumoniae* pKp199-2 in Canada in 2012 (accession no. [CP035537](#)). The structure of type I was *resD-orf-orf-orf-orf-orf-orf-IS26-bla*_{NDM-1}-*ble*_{MBL}-*bla*_{DHA-1}-*orf-ampR-hypA-sul1-ISCR1-ISEc28*, which was identical to the structures of plasmids of *K. pneumoniae* pMyNCGM084 in Myanmar in 2016 (accession no. [LC536683](#)). The structure of type J was *orf-pemI-pemK-IS1A-Tn3-IS26-IS30-bla*_{NDM-5}-*ble*_{MBL}-*trpF-tat-ISCR1-sul1-qacEΔ*, which was similar to that surrounding *bla*_{NDM-1} and identical to the structures of plasmids of *K. pneumoniae* p_b78b_NDM5 in Bangladesh in 2017 (accession no. [CP095588](#)). The structure of type K was *bla*_{TEM-1}-*rmtB-orf-groEL-IS26-IS30-bla*_{NDM-5}-*ble*_{MBL}-*trpF-tat-ISCR1-sul1-qacEΔ*, which was identical to the structures of plasmids of *E. coli* p1ESCUMp083_CORR in India in 2012 (accession no. [CP033159](#)). The structure of type L was *bla*_{TEM-1}-*rmtB-orf-groEL-IS26-IS30-bla*_{NDM-5}-*ble*_{MBL}-*trpF-tat-ISCR1-orf-orf-orf-pemK-pemI*, which was identical to the structures of plasmids of *E. coli* p_dm321b_NDM5 in Bangladesh in 2016 (accession no. [CP095607](#)). The structure of type M was *orf-bla*_{TEM-1}-*rmtB-orf-groEL-IS26-IS30-bla*_{NDM-5}-*ble*_{MBL}-*trpF-tat-ISCR1-IS26-Tn3-IS1B*, which was identical to the structure of plasmid of *K. pneumoniae* p_b199b_NDM5 in Bangladesh in 2017 (accession no. [CP095580](#)). The structure of type N was *orf-bla*_{OXA-181}-*orf-Tn3*, which was identical to the structure of plasmid of *K. pneumoniae* p47733_OXA_181 in Czech Republic in 2018 (accession no. [CP050368](#)). The structure of type O was *orf-mobD-orf-bla*_{OXA-232}-*rep-orf-mobC-mobB*, which was identical to the structure of plasmid of *K. pneumoniae* p_dm730a_OXA232 in Bangladesh in 2017 (accession no. [CP096175](#)). The structure of type P was *orf-mobC-orf-mobB-mobD-orf-bla*_{OXA-232}-*ereA-orf-orf-mobC-orf*, which was identical to the structure of plasmid of *K. pneumoniae* pColKP3 in India in 2017 (accession no. [CP036331](#)).

The isolates JUNP289 and JUNP328 harboured 3 and 2 copies of *bla*_{NDM-1}, respectively. These isolates had a copy of *bla*_{NDM-1} located on the same IncFII plasmids, and the surrounding structure was identical to that of type A (Fig. 3A). The remaining 2

copies of *bla*_{NDM-1} in JUNP289 were located on IncC/IncFIB(pNDM-Mar)/IncHI1B(pNDM-MAR) plasmid, and the surrounding structure was identified as type E and F (Fig. 3E and F), which were both unique. The remaining one copy of *bla*_{NDM-1} in JUNP328 was located on the IncC plasmid and the surrounding structure was identified as type B (Fig. 3B).

3.6. Genetic environments surrounding 16S rRNA methyltransferases *armA* and *rmtF*

The structures of the genomic environments surrounding *armA* and *rmtF* are shown in Fig. 4. The structure of type A was *sul1-IS91-IS5-armA-orf-orf-ISEc29-msr(E)-mph(E)*, which was detected in six of eight isolates and was identical to that of *K. pneumoniae* pKPN528-1 in the United States in 2012 (accession no. [CP020854](#)). The structure of type B was *orf-orf-orf-IS26-armA-orf-orf-ISEc29-msr(E)-mph(E)*, which was detected in one of eight isolates and was unique. The structure of type C was *ISCR1-IS5-armA-IS1-IS1-orf-orf-ISEc29*, which was detected in one of eight isolates and was identical to that of *K. pneumoniae* KP72 in China in 2013 (accession no. [CP101554](#)). The structure of type D was *int11-arr-2-itrA-aac(6')-Ib-cr-orf-rmtF-orf-ISC2-catB-gst-EAL-IS6100*, which was detected in 15 of 21 isolates and was identical to that of *K. pneumoniae* pBC16_IncFIA_IncFIK in India in 2021 (accession no. [CP122470](#)). The structure of type E was *arr-2-itrA-aac(6')-Ib-cr-orf-rmtF-orf-ISC2-catB-gst*, which was detected in 4 of 21 isolates and was unique. The structure of type F was *int11-arr-2-itrA-aac(6')-Ib-cr-rmtF-ISC2-catB-gst-EAL-IS6100*, which was detected in 2 of 21 isolates and was identical to that of *K. pneumoniae* p50595_NDM_1 in Czech Republic in 2019 (accession no. [CP050374](#)).

3.7. *bla*_{NDM-5} or *bla*_{OXA-181} in chromosomes

JUNP303, which belonged to ST101, had *bla*_{NDM-5}, *bla*_{CTX-M-15}, *rmtB*, and *aadA2* in its chromosome (Fig. 5A). The genetic structure surrounding *bla*_{NDM-5} was *IS6100-orf-bla*_{TEM-1}-*rmtB-orf-groEL-IS26-bla*_{NDM-5}-*ble*_{MBL}-*trpF-dsbD-ISCR1-sul1-qacEΔ-aadA2-dfrA12-Int11-IS26*. The structure of *IS26-bla*_{NDM-5}-*ble*_{MBL}-*trpF-dsbD-ISCR1-sul1-qacEΔ-aadA2-dfrA12-Int11-IS26* were identical to the structure

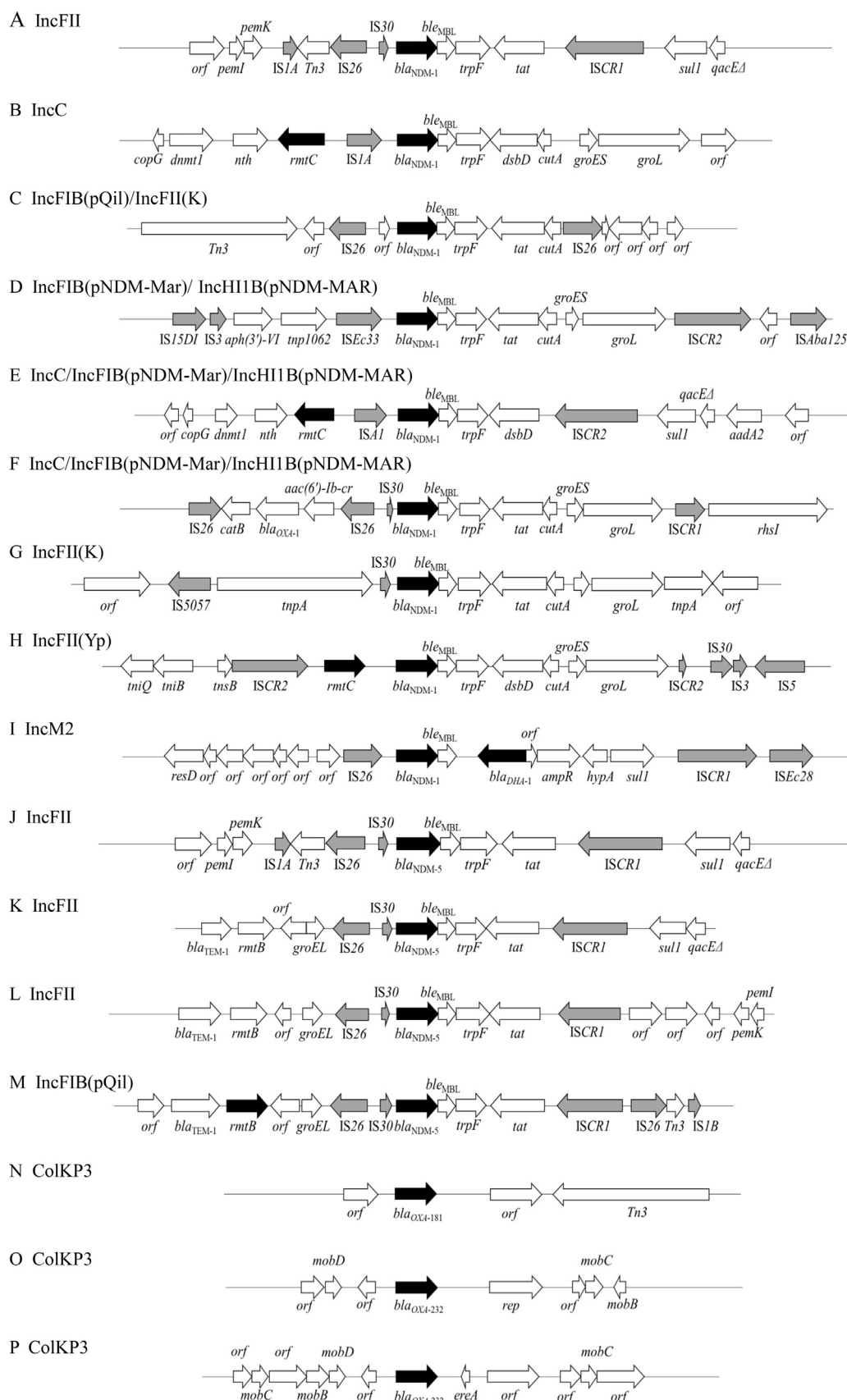


Fig. 3. Genomic environments surrounding *bla*_{NDM5} and *bla*_{OXA5} in *K. pneumoniae* isolated in Nepal. The structures of *bla*_{NDM-1} had 9 types (A-I), *bla*_{NDM-5} had 4 types (J-M), *bla*_{OXA-181} had 1 type (N), and *bla*_{OXA-232} had 2 types (O and P). The mobile genetic elements are shaded in gray and genes encoding carbapenemase and 16S rRNA methyltransferase are shaded in black.

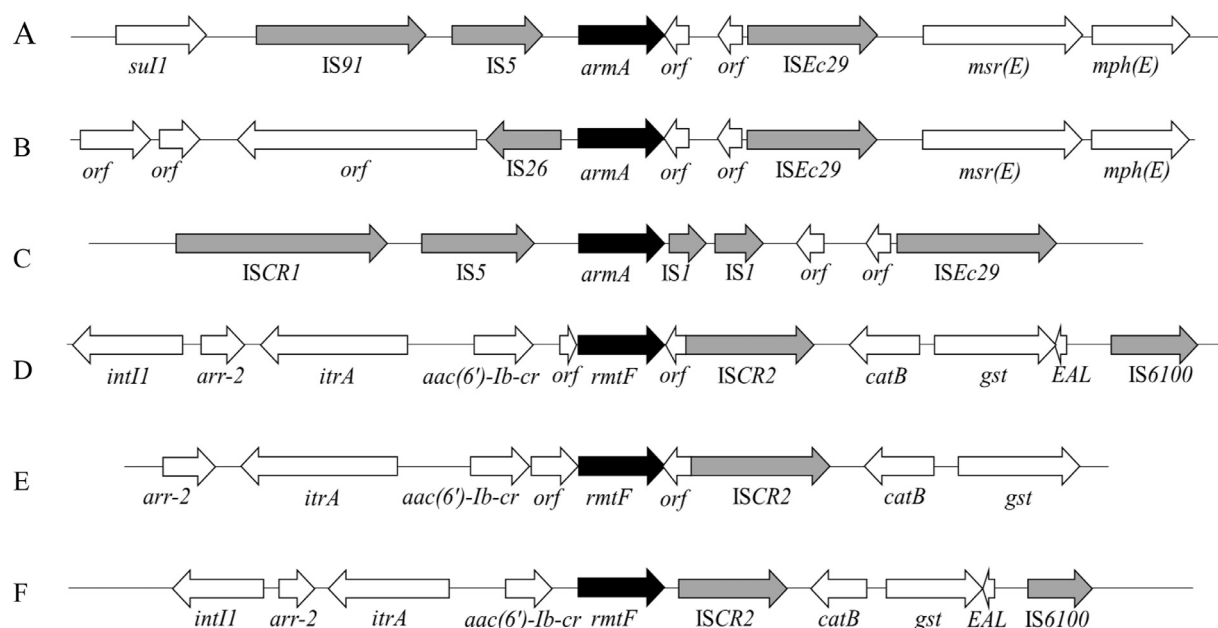


Fig. 4. Genomic environments surrounding *armA* and *rmtF* in *K. pneumoniae* isolate in Nepal. The structures of *armA* had 3 types (A–C) and *rmtF* had 3 types (D–F). The mobile genetic elements are shaded in gray and genes encoding carbapenemase and 16S rRNA methyltransferase are shaded in black.

of the plasmid of *E. coli* p1ESCUMp083_CORR in India in 2012 (accession no. CP034254).

Six of 13 isolates belonging to ST147 had 2 copies of *bla*_{OXA-181}, *bla*_{CTX-M-15}, *bla*_{SHV-11}, *rmtF* and *aac(6')-Ib-cr* in their chromosomes (Fig. 5B). The genetic structures surrounding *bla*_{OXA-181} were *ecpA*-*ecpB*-*orf*-*ISEcp1*-*bla*_{OXA-181}-*ecpD*-*orf*-*orf*-*gltJ*-*orf*-*orf* and 23S rRNA-5S rRNA-*ISEc1*-*bla*_{OXA-181}-*hdfR*-*orf*-*orf*-*orf*, which were both identical to that of *K. pneumoniae* CRKP-1215 chromosome in South Korea in 2014 (accession no. CP024838).

4. Discussion

K. pneumoniae ST147 and ST395 high-risk clones with high resistance to carbapenems and aminoglycosides are spreading in the medical settings across Nepal. In our previous study, molecular epidemiological analysis of 25 *K. pneumoniae* clinical isolates obtained in Nepal in 2012 revealed that 68% and 80% isolates were resistant to carbapenems and aminoglycosides, respectively, and most of the isolates belonged to clonal complex (CC) 11 (ST11, ST340, and ST437) and 14 (ST14 and ST15) [15]. These results indicate that the high-risk clones ST147 and ST395 show worldwide emergence and spread from 2012 to 2019 in medical settings across Nepal.

The variation of antimicrobial resistant factors associated with carbapenem-resistance seems to have increased from 2012 to 2019. The carbapenemase-encoding genes, including *bla*_{NDM-1}, *bla*_{NDM-5}, *bla*_{OXA-181}, and *bla*_{OXA-232}, were detected in isolates obtained in 2019, whereas *bla*_{NDM-1} and *bla*_{OXA-72} were detected in 2012. The *bla*_{NDM-5}, *bla*_{OXA-181}, and *bla*_{OXA-232} were initially reported in 2011 [16], 2011 [17] and 2013 [18], respectively. The review on *K. pneumoniae* producing NDMs described that the isolates harbouring *bla*_{NDM-1} were detected on a global scale, but *bla*_{NDM-5} were detected mostly in Asian countries [19]. On the other hand, *K. pneumoniae* harbouring *bla*_{OXA-181} and *bla*_{OXA-232} were detected on a global scale, but currently detected endemically in the Indian subcontinent [20]. In this study, the genomic structures surrounding *bla*_{NDM-1} are identical to that of the isolates from Bangladesh, Canada, Hong Kong, Myanmar, and the United States, but *bla*_{NDM-5}, *bla*_{OXA-181}, and *bla*_{OXA-232} were from Bangladesh, Czech Republic, and India. *K. pneumoniae* producing these carbapenemases in Nepal

will be influenced by the prevalence of these factors in neighbouring countries.

The high-risk clone isolates co-harboured *bla*_{NDM-5} and *rmtB* will likely increase in Nepal. When compared with the antimicrobial-resistant genes, no isolate in 2012 co-harboured *bla*_{NDM-5} and *rmtB* [15], whereas 57% of the isolates in 2019 co-harboured these two genes. On the other hand, 47% of the isolates in 2012 co-harboured *bla*_{NDM-1} and *armA* [15], whereas 8.7% of the isolates in 2019 co-harboured these two genes. The rate of *K. pneumoniae* isolates co-harboured *bla*_{NDM-1} and *armA* was 100% in Nepal in 2014 [21] and 58% (39/67 isolates) in India in 2015 [22]. These results indicate that carbapenemases and 16S rRNA methyltransferases drastically change in Nepal and its neighbouring countries.

IncFII plasmids co-carrying *bla*_{NDM-5} and *rmtB*, and ColKP3 plasmids co-carrying *bla*_{OXA-181}-₂₃₂ and *rmtF* will continue to spread in medical settings in Nepal. These IncFII and ColKP3 plasmids seem to be horizontally spreading in ST395 *K. pneumoniae* in Kathmandu in Nepal, as these isolates had plasmids with similar genetic structures. ST14, ST16, ST147, ST231, and ST395 *K. pneumoniae* co-harboured *bla*_{NDM-1}-₅ and *bla*_{OXA-181}-₂₃₂ have been reported in patients from India [23], Italy [24], Nepal [25], Singapore [26], South Korea [27], Thailand [28], the United Arab Emirates (UAE) [29] and the United States [30]. The clinical isolates of *K. pneumoniae* high-risk clones co-harboured *bla*_{NDM-1}-₅ and *bla*_{OXA-181}-₂₃₂ seem to emerge and spread worldwide.

We found that some *K. pneumoniae* isolates harboured multiple copies of carbapenemase- and 16S rRNA methyltransferase-encoding genes. Two ST15 *K. pneumoniae* isolates harboured 2 and 3 copies of *bla*_{NDM-1}, and three isolates ST15 and ST147 co-harboured *rmtB* and *rmtC/rmtF* on the same plasmids. In a previous study, the plasmids carrying 2 and 3 copies of *bla*_{NDM-1} were detected in *E. coli* in China in 2019 [31] and ST15 *K. pneumoniae* in Nepal in 2011 [21], respectively. Our present study indicates that copy numbers of *bla*_{NDM-1} were not affected by susceptibilities to carbapenems. On the other hand, 3 isolates producing NDM-1/-5 harboured two types of 16S rRNA methyltransferase-encoding genes, including *rmtB* and *rmtC/rmtF* on the same plasmids. In our previous study, *E. xiangfangensis* producing NDM-1 obtained from inpatients in Myanmar co-harboured two or three genes of 16S

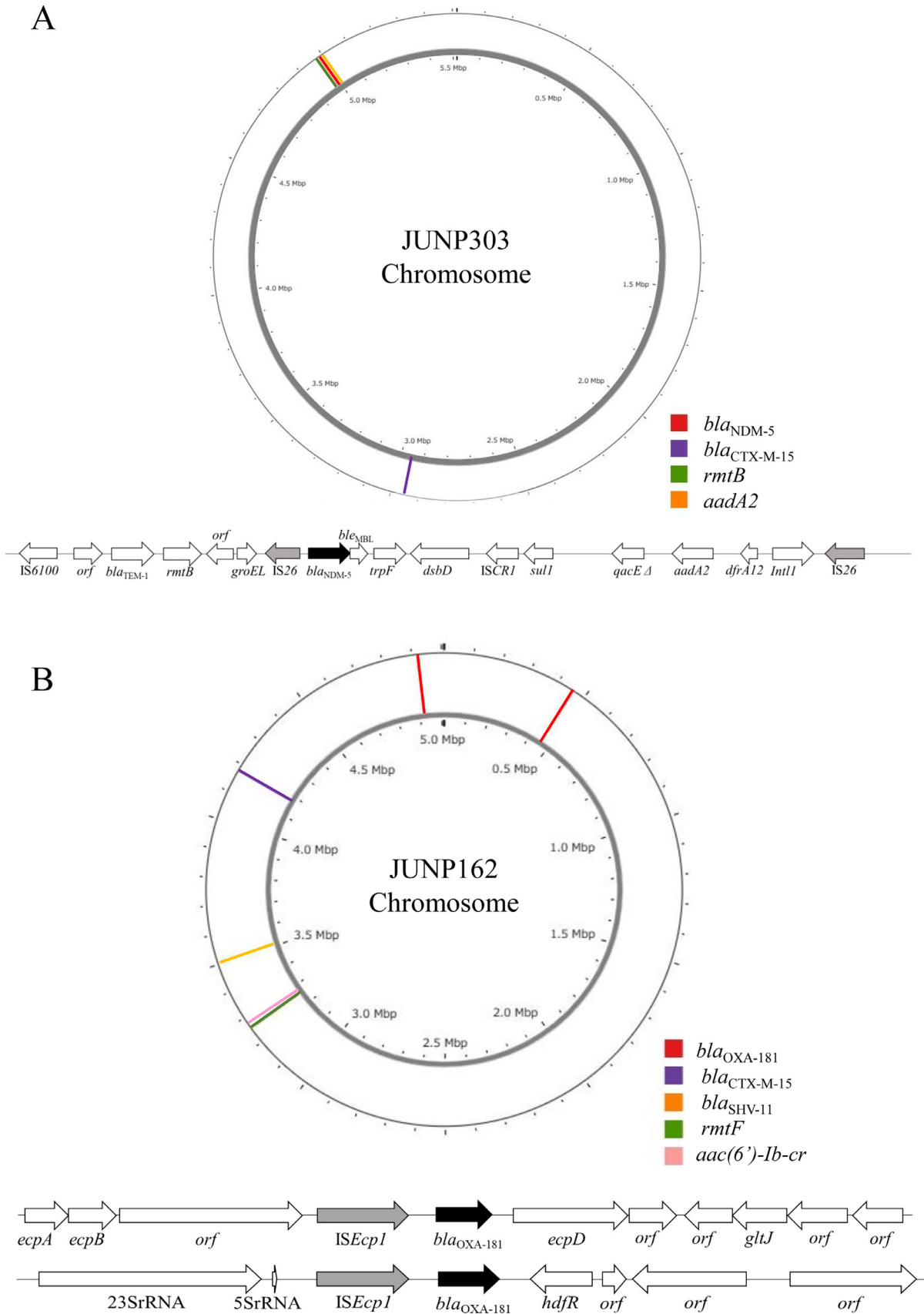


Fig. 5. The genomic environments surrounding *bla*_{NDM-5} and *bla*_{OXA-181} on chromosomes of JUNP303 and JUNP162. (A) JUNP303 harboured *bla*_{NDM-5}, *bla*_{CTX-M-15}, *rmtB* and *aadA2* on the chromosome. The *bla*_{NDM-5} was flanked by IS26. (B) JUNP162 harboured *bla*_{OXA-181}, *bla*_{CTX-M-15}, *bla*_{SHV-11}, *rmtB*, and *aac*(6')-Ib-cr on the chromosome. The *bla*_{OXA-181} was adjacent to ISEcp1. The images were generated using proksee (<https://proksee.ca/>).

rRNA methyltransferases, including *armA*, *rmtC*, and/or *rmtE* [32]. It is unclear the reason why *K. pneumoniae* and *E. xiangfangensis* had multi-copies of carbapenemase- and 16S rRNA methyltransferase-encoding genes.

Another key finding is a *K. pneumoniae* isolate with *bla*_{NDM-5} on the chromosome. In general, *Enterobacteriales* harboured carbapenemase-encoding genes, including *bla*_{NDMs} and *bla*_{OXA_s}, located on plasmids. Insertion sequence (IS) elements surrounding these genes will contribute to the integration of these genes into chromosomes. Sakamoto *et al.* reported that the chromosomal integration of *bla*_{NDM-5} occurred via IS26-mediated transposition in *E. coli* [33]. In this study, *bla*_{NDM-5} on the chromosome may have integrated with the same mechanism because the genetic structure was IS26-*bla*_{NDM-5}-*ble*_{MBL}-*trpF*-*dsbD*-*ISCR1*-*sul1*-*qacEΔ*-*aadA2*-*dfrA12*-*Int1*-IS26. The chromosomal integration of *bla*_{OXA-181} will be occurred via a one-ended transposition process by *ISEcp1* located upstream of *bla*_{OXA-181} [34]. Our isolates harbouring *bla*_{OXA-181} on the chromosome seem to be integrated by *ISEcp1* because of the presence of *ISEcp1* upstream of *bla*_{OXA-181}.

In conclusion, this is the first report describing the molecular epidemiology of multidrug-resistant *K. pneumoniae* clinical isolates in Nepal. The multidrug-resistant *K. pneumoniae* strains harboured various of genes encoding carbapenem- and aminoglycoside-resistant factors on plasmids or chromosomes. In particular, the specific plasmids IncFII and ColKP3 carrying *bla*_{NDMs}, *bla*_{OXA-48-like}s and genes encoding 16S rRNA methyltransferases horizontally spread in *K. pneumoniae* across this country. The spread of multidrug-resistant genes seems to be imported from neighbouring countries or modified domestically through plasmids and chromosomes. It is important to continue epidemiological surveillance of antimicrobial resistant *Enterobacteriales* in the medical settings of Nepal.

Declarations

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Competing interests: None of the authors has a conflict of interest.

Ethical approval: This study was approved by the Ethical Committee, Institute of Medicine, Tribhuvan University [approval number: 219 (6–11-E) /075/076], and the Biosafety Committee, Juntendo University (approval number: BSL2/29–1).

Author contributions: TT and ST designed the study and drafted the manuscript. JBS collected the clinical samples. TH and AK performed sequencing and analysed the data. YT, TM, and TK critically read and edited the manuscript. All authors read and approved the final manuscript.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jgar.2024.04.008.

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